LD4WAM: Learning Latent Dynamics from Human Videos for World Action Models

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Project page: https://stubborn111.github.io/LD4WAM/

Abstract

Human video is playing an increasingly central role in training World Action Models (WAMs), owing to its diversity and low collection cost relative to teleoperated robot data. However, most WAMs learn from such video only by predicting pixel-level future frames, giving dynamics that are not directly actionable, whereas motion retargeting recovers directly actionable actions but leaves a large visual gap across embodiments. We therefore propose motion-aligned latent dynamics as an embodiment-agnostic representation to bridge video priors and low-level actions. We further present LD4WAM, which pairs a Latent Dynamics Model trained with semantic reconstruction and real motion alignment with a World Dynamics Action Model built as a mixture-of-transformers (MoT), which preserves full future-video generation and uses learnable queries to distill these latent dynamics from generated futures for action conditioning. Pretrained on our curated unified dataset of over 5,000 hours of human and robot data, LD4WAM performs strongly in RoboTwin simulation and on real robots equipped with both grippers and dexterous hands, while generalizing well to unseen objects and backgrounds.

Introduction

World Action Models (WAMs), which unify future observation prediction with action generation conditioned on it, have recently emerged as a leading paradigm for robot manipulation (Li et al. 2026c; Ye et al. 2026; Yuan et al. 2026). Since prediction can be supervised with video alone, WAMs increasingly draw on large-scale egocentric human manipulation video, given its diversity and low collection cost relative to teleoperated robot demonstrations (Punamiya et al. 2026; Hoque et al. 2026; Grauman et al. 2022). However, many existing WAMs (Team et al. 2026; Bi et al. 2025a; Luo et al. 2026b) primarily use such video as pixel sequences supervised by future-frame reconstruction. As a result, the dynamics captured this way remain confined to the generative pathway and only weakly coupled to the policy, for lack of action labels to anchor them. This disconnect between prediction and action limits how effectively knowledge acquired from human video transfers to robot control.

A growing body of work therefore seeks to extract supervision from human video beyond pixel reconstruction. One line retargets human motion into the robot action space using hand or end-effector annotations from egocentric datasets (Zheng et al. 2026; Ma et al. 2026). However, such mappings can entangle transferable physical knowledge with embodiment-specific execution, introduce correspondence errors due to human-robot kinematic differences, and leave the visual embodiment gap unresolved. Another line learns latent actions from inter-frame pixel changes without action labels and uses them to condition robot control (Ye et al. 2025; Routray et al. 2025; Zhang et al. 2026). Yet these codes are designed to explain pixel changes rather than isolate motion, so they may retain appearance- and embodiment-specific information while lacking a clear connection to executable motion. Human video therefore requires an intermediate representation that abstracts away appearance and embodiment differences, transfers across embodiments, and is grounded in real motion to support robot control.

We propose *motion-aligned latent dynamics*, a compact representation of inter-frame dynamics learned in semantic feature space and explicitly aligned with observed motion. This representation provides an **embodiment-agnostic** bridge between high-dimensional video priors and low-level robot actions, enabling effective learning from both human and robot data. Semantic abstraction suppresses appearance- and embodiment-specific details, allowing the representation to capture dynamics shared across humans and robots. Motion alignment further grounds this representation in real end-effector motion, making it actionable for robot control.

Based on this, we present **LD4WAM**, which consists of two components (as shown in Fig. 1): 1. **Latent Dynamics Model (LDM)** first learns the bridge representation. In contrast to prior latent-action models that reconstruct pixels, the LDM is trained using next-frame *semantic reconstruction* together with *motion alignment*. These objectives encourage the latent space to capture motion-relevant semantic changes while suppressing appearance-specific variation. 2. **World**

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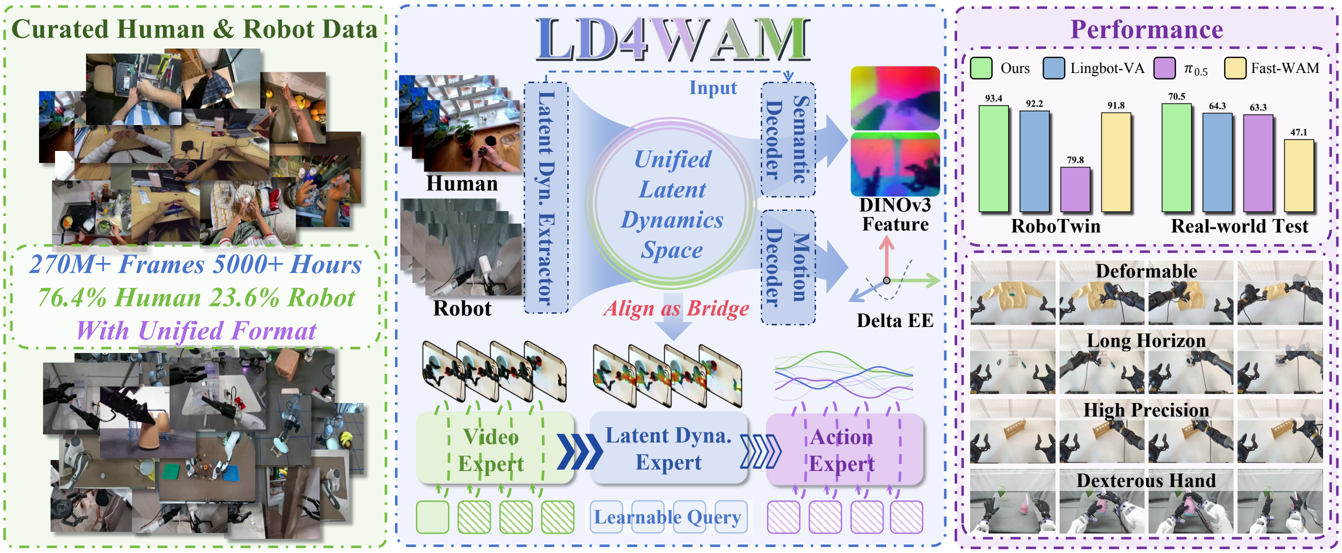

Figure 1: Overview of LD4WAM. Left: a unified pretraining dataset of 5,000+ hours human and robot videos. Middle: a Latent Dynamics Model distills motion-aligned latent dynamics via semantic and motion (Delta EE) supervision; in the World Dynamics Action Model, learnable queries carry these latent dynamics to bridge the video and action experts. Right: qualitative and quantitative results on RoboTwin and real robots demonstrate the strong manipulation capability of LD4WAM.

Dynamics Action Model (WDAM) then incorporates this representation while retaining full future-video generation. Learnable queries distill the LDM-defined latent dynamics from generated futures, and the resulting dynamics condition the action expert. The action expert therefore operates on a compact, motion-aligned dynamics signal rather than on raw pixels alone, while the model preserves the physical priors acquired by the pretrained video generator.

To enable pretraining at scale, we curate a corpus of human and robot manipulation data and process it through a unified cleaning pipeline. We filter out clips with excessive egocentric camera motion, hands outside the field of view, or insufficient manipulation motion. The remaining clips are standardized into a unified LeRobot format (Cadene et al. 2024), with end-effector deltas expressed in a shared camera coordinate frame as motion labels. The dataset contains approximately **270M frames**, corresponding to more than **5,000 hours** of video. Pretrained on this corpus, LD4WAM performs strongly on the RoboTwin simulation benchmark and in real-robot experiments using both bimanual two-finger grippers and dexterous hands. It handles long-horizon, high-precision, and dexterous manipulation tasks while generalizing well to unseen objects and backgrounds.

Our main contributions are summarized as follows:

- We introduce motion-aligned latent dynamics and a Latent Dynamics Model trained with semantic reconstruction and motion alignment, as an embodiment-agnostic bridge between video priors and low-level actions.
- We develop a World Dynamics Action Model that preserves full future-video generation and uses learnable queries to distill latent dynamics from generated futures for action conditioning.
- We curate a unified dataset comprising more than 5,000 hours of human and robot manipulation data and conduct extensive experiments in RoboTwin simulation and on real robots, demonstrating strong performance and generalization to unseen objects and backgrounds.

Related Work

Learning From Egocentric Human Videos

Egocentric human video is a scalable source of manipulation behavior for robot learning (Punamiya et al. 2026), exploited along two lines. The *mapping* line retargets human demonstrations into a shared action space and co-trains a policy on human and robot data (Zheng et al. 2026; Ma et al. 2026; Yang et al. 2025; Kareer et al. 2025; Li et al. 2026b); due to the embodiment gap it relies on tight viewpoint, speed, and kinematics alignment, and breaks down for grippers or dexterous hands. The *learning* line instead uses human video as a pretraining corpus for transferable visual or dynamics representations (Kareer et al. 2024; Bi et al. 2025b; Li et al. 2026a; Luo et al. 2026a; Routray et al. 2025; Zhang et al. 2026), but the representation is never grounded in real motion, leaving the policy to recover the mapping from representation to action on its own from limited robot data. We follow the learning line, but ground the latent dynamics in real motion, which makes them directly usable for action.

World Action Models for Robot Manipulation

World action models predict the future observation and produce the action within a single model. Most of them do so in pixel space: they build on a pretrained video generator, which supplies physical priors learned from internet-scale video (Ye et al. 2026; Li et al. 2026c; Yuan et al. 2026; Bi et al. 2025a). A second group moves the prediction target into richer feature spaces, using semantic feature (Lyu

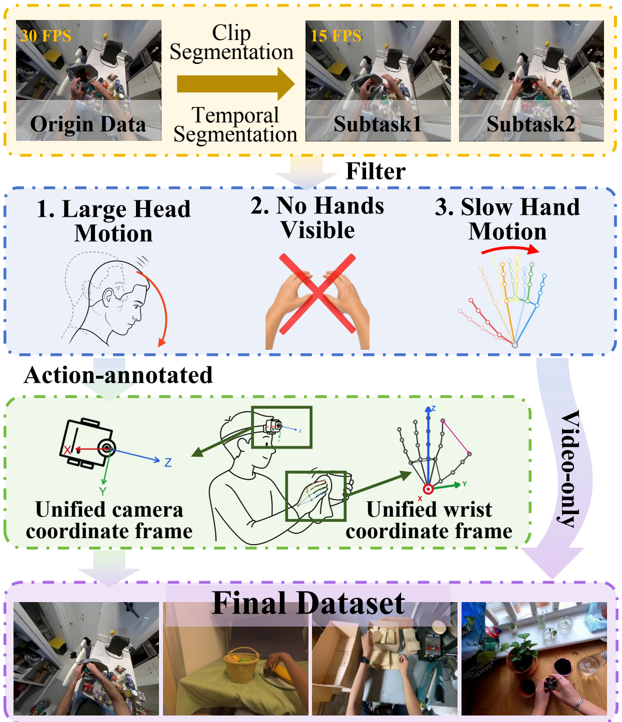

Figure 2: Human Data Process. Long videos are segmented into subtask clips, downsampled to 15 FPS, and filtered. Action-annotated clips are converted into unified camera and wrist coordinate frames, while video-only clips go directly to the final dataset.

et al. 2026; Tang et al. 2026; Luo et al. 2026b; Ma et al. 2026) or 3D structure information (Tian et al. 2026; Guo et al. 2026; Li et al. 2026d). However, these methods provide no representation that is both independent of embodiment and executable, limiting their use of unlabeled human video. LD4WAM learns such a representation in a semantic feature space, grounds it in real motion, and carries it to the action expert through learnable queries.

Method

As shown in Fig. 3, LD4WAM comprises a *Data Processing and Composition* pipeline and two models trained sequentially. Using the dataset curated by this pipeline, a *Latent Dynamics Model* (LDM) learns a motion-aligned latent dynamics space that bridges visual observations and low-level actions. A *World Dynamics Action Model* (WDAM) then predicts future observations while using these latent dynamics to ground action prediction. The complete *Training Procedure* is described at the end of this section.

Data Process and Composition

We process both human and robot datasets using the annotations provided by each dataset, and unify all sources into the LeRobot format. Long videos are segmented into subtask-level clips according to the annotated temporal boundaries. Clips shorter than 3 s or longer than 60 s are removed, and the remaining clips are uniformly downsampled to 15 FPS.

For the human datasets, we further apply the filtering pipeline in Fig. 2, handling two kinds of data. For *action-annotated* data with head-camera extrinsics and hand poses, we filter clips with large head motion (from the extrinsics), with both hands unannotated for an extended duration, or with little wrist motion over long windows; for each retained clip we express all actions in a unified head-camera coordinate frame and a unified wrist coordinate frame. For *video-only* data without action labels, we apply optical-flow camera-shake filtering together with MediaPipe (Lugaresi et al. 2019) to keep only steady segments with visible, actively manipulating hands. For the robot datasets, we perform the corresponding format conversion into the same organization. All detailed procedures are provided in Appendix A.

Finally, we aggregate eight datasets spanning egocentric human videos and multi-embodiment robot demonstrations (contributors 2024; Hou et al. 2026; Wu et al. 2025; LightwheelAI 2025; Hoque et al. 2026; Punamiya et al. 2026; Grauman et al. 2022; Ropedia 2026), and unify them into a single large-scale, cross-embodiment pretraining dataset. After the above process, the resulting dataset comprises 274.66M frames (5,086 hours) at 15 FPS, of which human and robot data account for 76.4% and 23.6% of the frames, respectively. The detailed composition and per-dataset statistics are reported in Appendix B.

Latent Dynamics Model

The LDM captures *what changes* between frames while discarding static appearance. Given a video, we sample a frame sequence $\{o_t, o_{t+n}, \ldots, o_{t+L}\}$ with a skip n drawn from a set of temporal strides, which exposes the model to transitions at varying temporal scales rather than a fixed frame rate. Each frame is encoded by a frozen DINOv3 (Siméoni et al. 2025) into semantic features f_i; a spatio-temporal transformer then aggregates them across space and time, and a *soft vector quantization* (Chen et al. 2025a) layer maps each transition into a latent-dynamics token z_i, whose soft assignment to the codebook keeps the space smooth and structured while suppressing low-level visual noise. Two heads supervise z_i during training: a *semantic decoder* takes the current-frame feature f_i together with z_i and reconstructs the future features $\hat{f}_{i+n}$, forcing z_i to encode the semantic evolution from f_i rather than appearance; a *motion decoder* predicts the inter-frame motion $\hat{m}_i$ as a delta end-effector pose, aligning the dynamics with real motion. The objective is

$$\mathcal{L}_{\text{LDM}} = \mathcal{L}_{\text{sem}} + \lambda_m \mathcal{L}_{\text{mot}} + \mathcal{L}_{\text{vq}}, \tag{1}$$

$$\mathcal{L}_{\text{vq}} = \lambda_u \mathcal{L}_{\text{use}} + \lambda_e \mathcal{L}_{\text{ent}}, \tag{2}$$

where $\mathcal{L}_{\text{sem}}$ is a cosine distance in the frozen DINOv3 feature space augmented by a small feature-norm matching term; $\mathcal{L}_{\text{mot}}$ is applied only to frames that carry action labels, so all video shapes the semantic dynamics while action-labeled data additionally grounds them in executable motion; and $\mathcal{L}_{\text{vq}}$ is a codebook regularizer comprising the KL divergence from the mean code-assignment distribution to a uniform prior ($\mathcal{L}_{\text{use}}$) and the mean per-token soft-assignment entropy ($\mathcal{L}_{\text{ent}}$), jointly preventing codebook collapse.

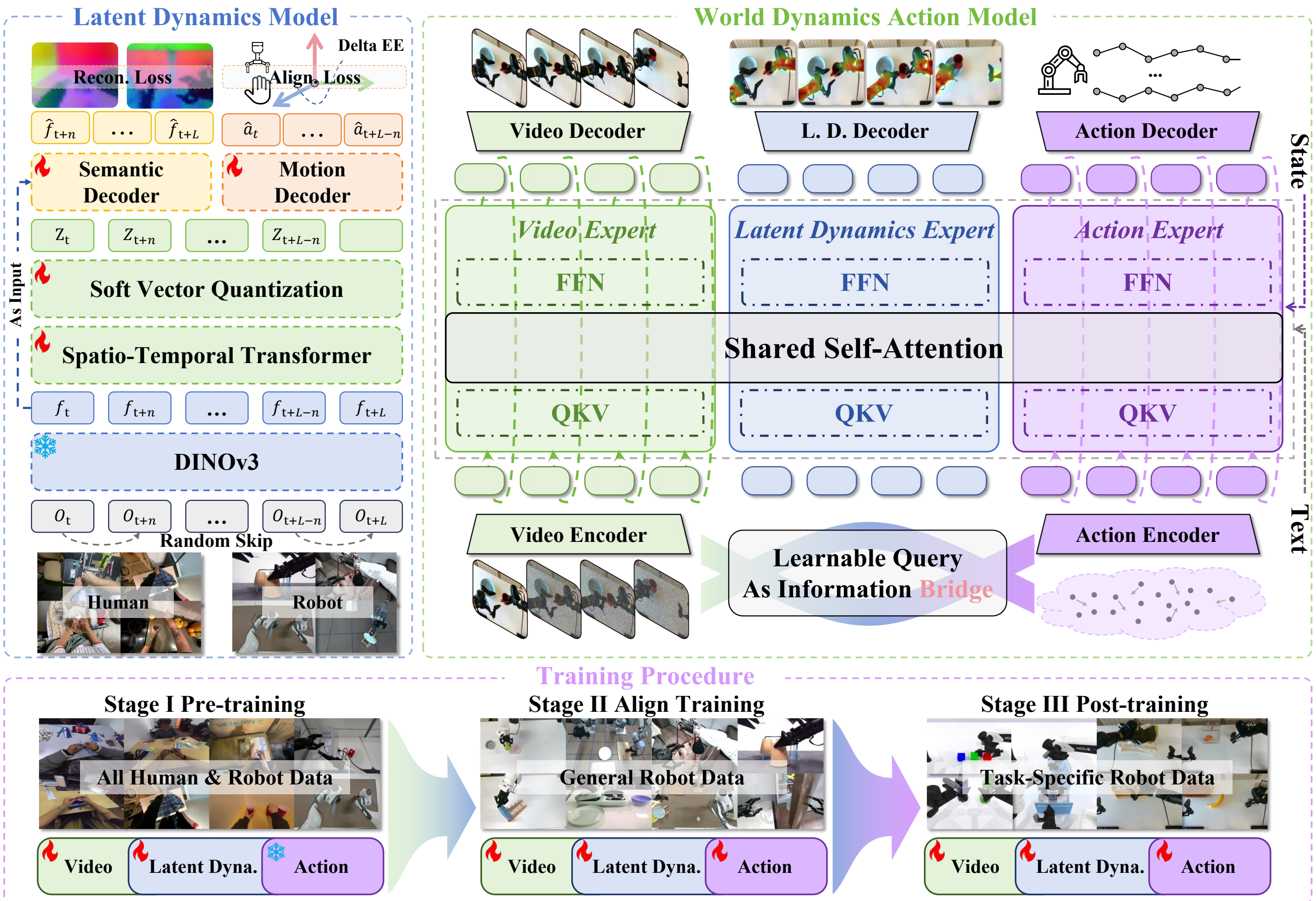

Figure 3: Pipeline of LD4WAM. Latent Dynamics Model (left): Frames sampled with random temporal skips are encoded by a frozen DINOv3; their features are aggregated by a spatio-temporal transformer and mapped via soft-VQ to latent dynamics $\{z_i\}$, supervised by semantic reconstruction and camera-frame end-effector deltas (Delta EE). World Dynamics Action Model (right): A Mixture-of-Transformers in which the video, latent dynamics, and action experts share a self-attention module. Learnable queries transfer the LDM-supervised latent dynamics from the video expert to the action expert; text conditions all three experts through cross-attention and is additionally concatenated with the state as input to the action expert. Training Procedure (bottom): Stage I pretrains the video and latent dynamics experts on all data while freezing the action expert; Stage II aligns all experts on robot data; Stage III post-trains the model on task-specific data.

At inference, we retain only the encoding path: given a frame sequence of the training length, DINOv3, the spatio-temporal transformer, and the SoftVQ layer produce the latent-dynamics tokens $\{z_i\}$, while both decoders are discarded. These tokens are precomputed on the training videos and serve as the prediction targets that supervise the WDAM's latent dynamics expert.

World Dynamics Action Model

WDAM is a Mixture-of-Transformers architecture whose three experts keep separate weights and interact only through a shared self-attention over the concatenated token sequence. A *video expert* and an *action expert* are flow-matching denoisers that generate the future video and the future action, respectively; the video expert is initialized from a pretrained video generation model Wan2.2 (Wan et al. 2025) to inherit its physical priors. Between them sits a *latent dynamic expert*: a clean stream of learnable queries, with no noise or diffusion timestep, trained to reproduce the future latent dynamics under supervision from the frozen LDM.

What turns these latent dynamics into an effective bridge is the asymmetric visibility of the shared attention, which makes the layer stack a one-directional pipeline video $\rightarrow$ latent dynamics $\rightarrow$ action. The video stream attends only to itself, so future-video generation stays self-contained and the pretrained prior is preserved unchanged, never perturbed by the downstream streams. The learnable queries attend only to the predicted video and themselves, distilling *what will happen* into a compact summary while staying blind to the action stream. The action stream then attends to both the predicted video and this summary at every layer, so control is grounded on the purified latent dynamics rather than left to read raw pixels alone. At inference this reduces to two stages: the video is denoised first, and the action is then solved from the frozen video together with the one-shot latent dynamics, so the bridge adds no extra denoising steps.

For training, the three experts are supervised jointly,

$$\mathcal{L}_{\text{WDAM}} = \lambda_v \mathcal{L}^{\text{fm}}_{\text{video}} + \lambda_d \mathcal{L}^{\text{mse}}_{\text{dyn}} + \lambda_a \mathcal{L}^{\text{fm}}_{\text{act}}, \tag{3}$$

where $\mathcal{L}^{\text{fm}}_{\text{video}}$ and $\mathcal{L}^{\text{fm}}_{\text{act}}$ are flow-matching losses for the video and action denoisers, and $\mathcal{L}^{\text{mse}}_{\text{dyn}}$ is a masked MSE between the predicted latent dynamics and the offline LDM targets $\{z_i\}$. The loss weights λ_v, λ_d, and λ_a are varied across the three training stages to reflect their shifting objectives.

Training Procedure

We train the LDM first on a high-quality subset obtained through stricter filtering and freeze it, then optimize the WDAM in three stages. **Stage I (Pre-training)** trains the video and latent dynamics experts on all human and robot data while freezing the action expert, which lacks labels on human video; this teaches the model to imagine the future and how to extract latent dynamics from it at scale. **Stage II (Align Training)** unfreezes all experts and trains on general robot data, aligning the predicted latent dynamics with executable actions. **Stage III (Post-training)** fine-tunes all experts on task-specific robot data for deployment.

Throughout, the DINOv3 encoder stays frozen and the video expert is initialized from a pretrained Wan2.2 generator. Full model dimensions, optimizer, learning rates, and per-stage schedules are provided in Appendix C.

Experiment

Setup

Simulation. We evaluate on the RoboTwin benchmark (Chen et al. 2025b) extensively using its standard suite of 50 tasks. For each task we train on standard 50 clean and 500 randomized demonstrations, and train a single unified policy across all tasks. Each task is evaluated over 100 rollouts, and we report the success rate. We compare LD4WAM against state-of-the-art VLA and WAM methods, as well as baselines that leverage large-scale human data.

Real World. We deploy on two embodiments: a dual-arm PIPER equipped with parallel grippers, and a dual-arm Tianji platform equipped with Wuji dexterous hands. On these we build five gripper tasks and two dexterous-hand tasks that span a range of manipulation skills: *Sorting* (long-horizon pick-and-place by category), *Shift Test Tube* (high-precision insertion), *Tidy Desk* (long-horizon multi-step tidying), *Fold Shirt* (deformable cloth manipulation), and *Handover Mug* (bimanual cross-arm coordination), together with two dexterous-hand tasks, *Place Rubik's Cube* (multi-finger grasp and placement) and *Spray Water* (in-hand tool use requiring independent finger actuation to operate a trigger). To probe generalization, we further add *object* and *background* generalization settings on Tidy Desk, Fold Shirt, and Handover Mug, evaluating on unseen objects and unseen backgrounds; representative setups for both settings are shown in Fig. 4. We collect 50 demonstrations per task and evaluate each setting over 30 rollouts. Further task details are in Appendix D. For baselines, we compare against $\pi_{0.5}$ (Intelligence et al. 2025), a state-of-the-art VLA, and Fast-WAM (Yuan et al. 2026) and Lingbot-VA (Li et al. 2026c), two leading world action models.

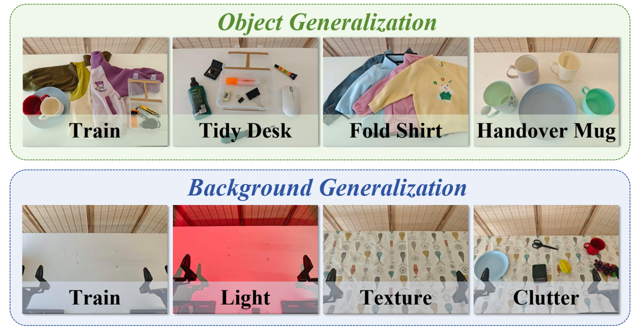

Figure 4: Generalization Setting. We evaluate generalization along two axes: object and background. Object Generalization (top) replaces the training objects with unseen ones for three tasks. Background Generalization (bottom) perturbs the visual scene while keeping the task fixed, via lighting, table texture, and distractor objects.

Table 1: Performance comparison on RoboTwin.

Method	Clean	Random	Average
$\pi_{0.5}$ (Intelligence et al. 2025)	82.74	76.76	79.8
ACE-Ego-0 (Li et al. 2026b)	91.12	90.62	90.9
Motus (Bi et al. 2025a)	88.66	87.02	87.8
Lingbot-VA (Li et al. 2026c)	92.90	91.50	92.2
Fast-WAM (Yuan et al. 2026)	91.88	91.78	91.8
Being-H0.7 (Luo et al. 2026b)	90.20	89.60	89.9
LaWAM (Chen et al. 2026)	92.64	89.80	91.2
Ours	**93.96**	**92.78**	**93.4**

Latent Dynamics Regression. Because the latent dynamics cannot be scored directly by task success, we measure how much action-relevant information they carry by regressing the end-effector motion from them. We freeze the LDM, extract latent dynamics from video clips, and train a shallow head to read out the per-step end-effector delta; being shallow, it probes what the latent already encodes rather than learning new features, and we report the scale-normalized MSE. We build the evaluation set from RoboTwin over all 50 tasks, using 2,500 clean and 2,500 randomized episodes, cut into non-overlapping 8-frame clips split 8:2 into probe-training and validation. To test temporal robustness, we sample the clips at several frame strides and group them into a *high-rate* band (12.5–50 Hz) and a *low-rate* band (2.5–10 Hz), which probe regression over small and large motions, respectively. For each stride we train a separate head under the same configuration; further details are in Appendix E.

Result Analysis

Latent Dynamics Retrieval. As shown in Fig. 5, we retrieve the nearest neighbors of a two-frame query by cosine similarity of latent dynamics across different human and robot datasets; the query is a robot arm under opposite

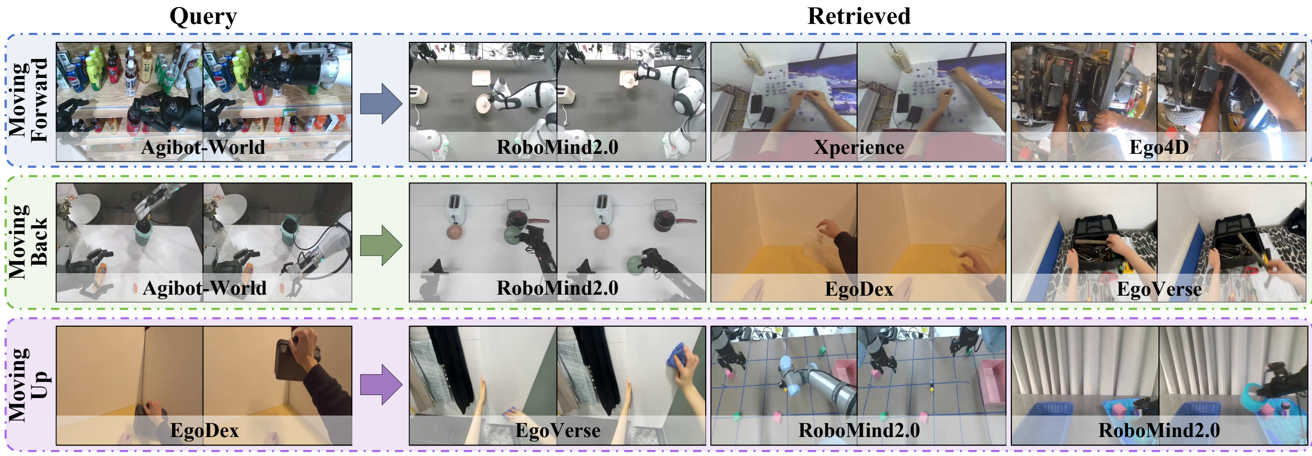

Figure 5: Cross-domain latent dynamics retrieval. Nearest neighbors of each query clip (left) by cosine similarity in the frozen LDM latent space, pooled across human and robot data; the source dataset is labeled under each panel.

Table 2: Success rate (%) on real-world gripper and dexterous-hand manipulation tasks.

	Gripper					Dexterous Hand		
Method	Sorting	Shift Test Tube	Tidy Desk	Fold Shirt	Handover Mug	Place Rubik's Cube	Spray Water	Average
$\pi_{0.5}$	63.3	43.3	**73.3**	76.7	70.0	76.7	40.0	63.3
Fast-WAM	50.0	10.0	53.3	46.7	63.3	76.7	30.0	47.1
Lingbot-VA	**80.0**	40.0	63.3	73.3	76.7	80.0	36.7	64.3
Ours	**80.0**	**50.0**	70.0	**80.0**	**83.3**	**83.3**	**46.7**	**70.5**

motions in rows 1 and 2 and a human hand in row 3. The retrieved neighbors consistently reproduce the query's motion, and the two opposite queries retrieve neighbors that move in correspondingly different ways. This pattern holds not only across robot embodiments but even between a human hand and a robot arm sharing the consistent latent dynamics.

Simulation. As shown in Tab. 1, LD4WAM reaches an average success rate of 93.4% across the 50 tasks on the RoboTwin benchmark, outperforming the strong VLA baseline ACE-Ego-0, which is also trained on human data, by $+2.5$ points and the leading WAM Lingbot-VA by $+1.2$ points. The top methods are tightly clustered in simulation, so we treat this benchmark as a competitiveness check and defer the discriminative comparison to the real world.

Real World. As shown in Tab. 2, LD4WAM attains the best success rate on six of the seven real-world tasks across both embodiments, raising the average from 63.3%, 47.1%, and 64.3% for $\pi_{0.5}$, Fast-WAM, and Lingbot-VA to 70.5%. Fig. 6 further provides qualitative rollouts on each task.

Generalization. Beyond the in-distribution comparison, Tab. 4 evaluates both generalization settings. On unseen objects, LD4WAM retains 88.6% of its in-distribution performance, the highest of all methods, confirming that motion-aligned latent dynamics are largely invariant to *which* object is manipulated. Background shift is more revealing: although LD4WAM improves over prior world action models by $+10.0$ and $+34.4$ points, it still trails $\pi_{0.5}$ (44.4% vs. 54.5%), with the gap concentrated on texture-level perturbations. We attribute this to a structural property of the world-action paradigm: control is conditioned on generated future video, and an unseen texture is precisely what video prediction handles worst, so the action expert must act on a degraded rollout, whereas a VLA inherits texture-invariant features from an internet-scale backbone. The latent dynamics mitigate this by discarding much of the appearance variation in a frozen semantic feature space, but they are read out from the same generated video, so the effect is suppressed rather than removed. Appendix G provides additional qualitative rollouts under object and background shifts.

Table 3: Ablation of LDM under latent dynamics regression (scale-normalized MSE $\downarrow$, averaged within each band).

Rate band	**LDM**	w/o motion align.	w/o multi-stride	w/ pixel recon.	w/ hard VQ
High-rate	**0.21**	0.78	0.24	0.25	0.37
Low-rate	**2.13**	10.59	2.54	2.69	4.00

Ablation Analysis

Latent Dynamics Model Ablation. Tab. 3 ablates the four LDM ingredients under the latent dynamics regression. Motion alignment is the key one, tying the latent dynamics to real end-effector motion so they track body motion rather than background or appearance change; removing it inflates the error by 3.7–5.0×, by far the largest effect. Soft VQ keeps the space continuous to match continuous motion, and

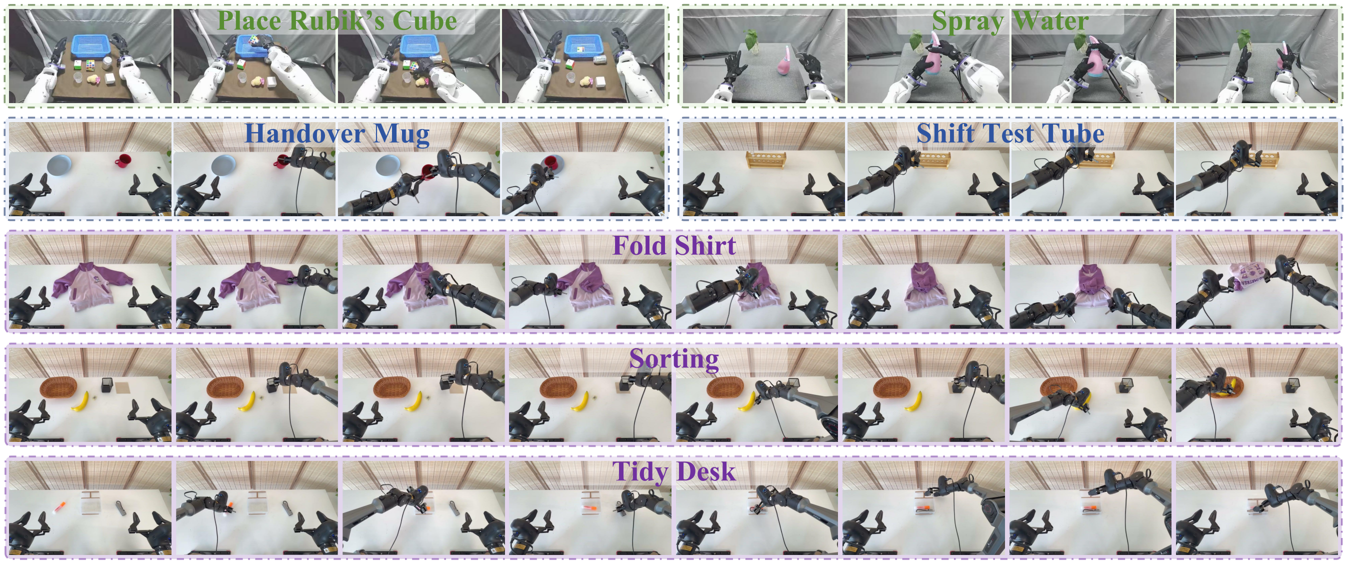

Figure 6: Real-world Execution. Qualitative real-world rollouts of our policy across all evaluation tasks, spanning both dexterous-hand and gripper manipulation.

Table 4: Generalization and architecture ablations (success rate, %). "Obj"/"Bg" denote object/background generalization. Bold marks the best result per column. "+" cumulatively adds a component that is retained thereafter; "↪" replaces the component added immediately above with an alternative that is neither stacked nor carried forward. The last row is the full model.

Method	Tidy Desk			Fold Shirt			Handover Mug			Average
	Base	Obj	Bg	Base	Obj	Bg	Base	Obj	Bg	
$\pi_{0.5}$	**73.3**	53.3	**56.7**	76.7	60.0	**50.0**	70.0	66.7	**56.7**	62.6
Fast-WAM	53.3	30.0	13.3	46.7	23.3	6.67	63.3	33.3	10.0	31.1
Lingbot-VA	63.3	43.3	30.0	73.3	63.3	33.3	76.7	70.0	40.0	54.8
Ablation of our architecture										
Video-Action Dual-Expert WAM	56.7	30.0	13.3	63.3	56.7	10.0	66.7	50.0	13.3	40.0
+ Latent Dynamic Expert (Ours)	63.3	40.0	20.0	73.3	63.3	16.7	73.3	63.3	20.0	48.1
↪ LDM w/o motion alignment	56.7	36.7	16.7	70.0	60.0	13.3	66.7	56.7	13.3	43.3
↪ LDM w/ pixel reconstruction	60.0	36.7	13.3	66.7	63.3	16.7	66.7	60.0	16.7	44.5
+ Pre-training	63.3	53.3	40.0	76.7	66.7	40.0	73.3	70.0	46.7	58.9
+ Align Training (Full)	70.0	**56.7**	43.3	**80.0**	**70.0**	40.0	**83.3**	**80.0**	50.0	**63.7**

hard quantization still adds 76–88%. Semantic reconstruction keeps the latent dynamics on high-level content, whereas reconstructing pixels ties them to noisy low-level appearance. Multi-stride sampling covers both large and small inter-frame motions, whereas a single scale covers only a narrow range.

Full Architecture Ablation. Tab. 4 ablates the full architecture by real-world success rate, building up from a Video-Action Dual-Expert WAM baseline (40.0%). Adding the Latent Dynamic Expert raises the average success rate to 48.1%. Its benefit hinges on how the LDM targets are supervised, and ablating either of the two signals that shape them degrades control: removing motion alignment, so the latent dynamics are never grounded in executable motion, drops the average to 43.3%, while replacing the semantic feature-space objective with pixel reconstruction drops it to 44.5%. Pretraining adds a further 10.8 points, and is almost entirely a generalization gain: the in-distribution average moves by only 1.1 points, while unseen objects improve by 7.8 points and unseen backgrounds by 23.3 points, indicating that it supplies the visual and dynamic coverage the task-specific robot data lacks. Finally, the alignment middle-training stage yields the full model at 63.7%, adding 4.8 points overall; here the gains concentrate on execution quality and object generalization, showing that the two stages are complementary, alignment sharpens how the latent dynamics are consumed by the action expert, pretraining broadens what those dynamics have seen. Together they lift the baseline by 23.7 points.

Conclusion

We presented LD4WAM, a world action model that turns large-scale human and robot video into transferable, generalizable manipulation capabilities through motion-aligned latent dynamics. We first curate a unified dataset of over 5,000 hours of human and robot data into a shared format. On this dataset, a Latent Dynamics Model distills this representation via semantic reconstruction and real-motion alignment, and a

World Dynamics Action Model uses learnable queries to read it out as a signal that conditions action. Across RoboTwin and two real-robot embodiments, LD4WAM outperforms strong VLA and world action model baselines, with strong generalization to unseen objects and backgrounds.

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Appendix

A. Data Processing Details

Segmentation and temporal filtering. All source videos are segmented into subtask-level clips using the temporal boundaries provided by each dataset's annotations. We discard clips shorter than 3 s or longer than 60 s, and uniformly downsample the remainder from their native frame rate to 15 FPS.

Unified coordinate frames. For action-annotated human data, all actions of a clip are expressed relative to the head camera at the first frame, giving a *unified camera coordinate frame* that removes absolute pose and makes trajectories comparable across clips and datasets. In parallel, each hand is mapped to an end-effector in a *unified wrist coordinate frame*. Let p_w, p_m, p_i, and p_l be the keypoints of the wrist, the middle-finger knuckle, the index knuckle, and the little-finger knuckle, and let $\widehat{v} = v/\|v\|$. The frame is defined as follows:

- The origin is placed at p_w.
- $z = \widehat{p_m - p_w}$, pointing from the wrist along the hand.
- $x = z \times \widehat{(p_l - p_i)}$, the palmar normal, with the sign of $(p_l - p_i)$ taken per hand so that x always points from the back of the hand toward the palm.
- $y = z \times x$, along the knuckle line.

For the robot data, the same construction is applied to the gripper. Let q_b be the point at which the gripper attaches to the arm, q_t the midpoint between the fingertips, and q_1, q_2 the two jaw positions. The frame is defined as follows:

- The origin is placed at q_b.
- $z = \widehat{q_t - q_b}$, pointing from the base along the gripper.
- $x = z \times \widehat{(q_2 - q_1)}$, normal to the plane in which the jaws travel, with the labelling of q_1 and q_2 fixed per embodiment so that x agrees with the end-effector frame declared by the robot and is the same for both arms.
- $y = z \times x$, along the jaw opening direction.

The two constructions assign the same physical meaning to each axis. The exported motiond is the frame-to-frame difference of this pose.

Filtering criteria. For action-annotated data, we remove a clip if (i) the head-camera extrinsics indicate large head motion, (ii) both hands lack valid annotations for an extended duration, or (iii) the wrist moves little over a long temporal window. For video-only data, which has no extrinsics or action labels, we run MediaPipe hand detection and keep only segments in which at least 50% of frames contain a detected hand and the wrist trajectory spans more than 8 px. We additionally compute sparse Lucas–Kanade optical flow (Shi–Tomasi corners, 3-level pyramid) as a proxy for camera motion, and discard them if the mean per-frame flow magnitude over the clip exceeds 20 px (at the 15 fps sampling rate) or its range exceeds 80 px.

Table 5: Pre-training Dataset Statistics (FPS = 15).

Category	Dataset	Frames (M)	Hours
Human	Ego4D (Grauman et al. 2022)	99.88	1849.7
	EgoVerse (Punamiya et al. 2026)	50.96	943.6
	EgoDex (Hoque et al. 2026)	35.67	660.6
	EgoSuite (LightwheelAI 2025)	16.45	304.7
	Xperience (Ropedia 2026)	6.93	128.2
	Subtotal	**209.89**	**3886.9**
Robot	Agibot-World (contributors 2024)	30.23	559.8
	RM2.0-Franka (Hou et al. 2026)	21.00	388.9
	RM2.0-Agilex (Hou et al. 2026)	6.92	128.2
	RM2.0-UR (Hou et al. 2026)	3.88	71.8
	RM1.0-Franka (Wu et al. 2025)	2.73	50.6
	Subtotal	**64.77**	**1199.4**
Total		**274.66**	**5086.3**
Frame ratio		H 76.4% / R 23.6%	

B. Dataset Details

Tab. 5 lists the pre-training datasets: five egocentric human and three multi-embodiment robot datasets, unified into the LeRobot format at 15 FPS, with human data forming the majority of the corpus.

C. Training Details

Latent Dynamics Model. The spatial encoder is a frozen DINOv3 ViT-L/16, producing 196 patch tokens of dimension 1024 per frame. The spatio-temporal transformer has 6 non-causal layers (encoder) and 6 causal layers (decoder), each with 16 attention heads and head dimension 64. The SoftVQ codebook contains 8 entries; each transition is quantized into a 4×4 grid of 16 code slots, each of dimension 32, yielding a 512-dimensional latent vector per transition.

We find that large motion noise degrades the LDM training, so we tighten the thresholds of our data filtering pipeline (App. A) and re-filter the corpus into a cleaner subset of roughly 1,500 hours to train the LDM for semantic reconstruction. Motion alignment is applied to the four datasets with verified per-step delta end-effector labels (EgoVerse, EgoDex, Xperience, and Agibot-World).

Motion alignment is introduced after a warmup phase to allow semantic dynamics to stabilize before grounding. The model is trained with AdamW (lr=3×10^{-5}, cosine decay, weight decay 0.01, gradient clip 1.0).

World Dynamics Action Model. Our WDAM is a three-expert Mixture-of-Transformers built on a **Wan2.2 TI2V-5B** video-diffusion backbone. The video expert (a video DiT) has 30 layers, hidden dimension 3072, 24 attention heads of head dimension 128, and FFN dimension 14336, operating on the 16-channel latents of the Wan2.2 causal VAE ($4\times$ temporal compression); the T5 text encoder and the VAE are kept frozen, so only the video DiT is trained. Alongside it run two lightweight experts fused into the video expert through per-layer mixed self-attention (1:1 layer participation across all 30 layers, with the attention geometry, i.e., number of heads and head dimension, injected from the video backbone): (i) the **Latent Dynamics Expert**, a learnable-query

Hyperparameter	Value
Optimizer	AdamW (β_1, β_2)=$(0.9, 0.95)$
Weight decay	0.01
Gradient clip	1.0 (global norm)
LR schedule	cosine, 5% warmup
Peak learning rate	1×10^{-4}
Min learning rate	1% of peak
Precision	`bf16`
Sharding	DeepSpeed ZeRO-2
Grad. checkpointing	enabled
Seed	42
Frozen modules	T5 text encoder, VAE, LDM
Stage I (Pre-training)	
Trainable experts	video, latent dynamics
Views	1 (head)
Batch size / GPU	128
Schedule	1 epoch
$(\lambda_v, \lambda_d, \lambda_a)$	(1.0, 1.0, 0.0)
Stage II (Align Training)	
Trainable experts	video, latent dynamics, action
Views	3 (head + wrists)
Batch size / GPU	96
Schedule	1 epoch
$(\lambda_v, \lambda_d, \lambda_a)$	(1.0, 0.5, 1.0)
Stage III (Post-training)	
Trainable experts	video, latent dynamics, action
Views	target embodiment
Batch size / GPU	48
Schedule	5 epochs
$(\lambda_v, \lambda_d, \lambda_a)$	(1.0, 0.1, 1.0)

Table 6: Training hyperparameters of the World Dynamics Action Model (WDAM).

Figure 7: Real-robot platforms. Top: the gripper platform, a dual-arm AgileX PiPER with Pika Grippers. Bottom: the dexterous platform, dual Tianji arms with Wuji hands.

expert (residual width 512, FFN 2048) that emits k_d=4 latent-dynamics tokens per predicted video latent (2 predicted latents $\Rightarrow$ 8 tokens), regressing the 512-dimensional LDM target z_i; and (ii) the **Action Expert** (residual width 1024, FFN 4096) that flow-matches the robot action, predicting 16 actions per latent. The Action Expert shares a single transformer body across heterogeneous embodiments—covering both parallel-gripper and dexterous-hand robots—and adapts to each by swapping only its input and output projection layers, so the shared backbone need not commit to a fixed action dimension. The latent-dynamics target $\{z_i\}$ is produced by the frozen LDM run on the raw observation frames (online during Stage I, precomputed offline thereafter); at inference the LDM is not invoked and the Latent Dynamics Expert predicts $\{z_i\}$ directly.

The WDAM is trained under an inverse-dynamics protocol: video denoising is teacher-forced during training, while at inference the action is solved from the predicted video. The three experts are supervised jointly by

$$\mathcal{L}_{\text{WDAM}} = \lambda_v \, \mathcal{L}^{\text{fm}}_{\text{video}} + \lambda_d \, \mathcal{L}^{\text{mse}}_{\text{dyn}} + \lambda_a \, \mathcal{L}^{\text{fm}}_{\text{act}}. \tag{3}$$

The three-stage curriculum shifts the weights $(\lambda_v, \lambda_d, \lambda_a)$ to reflect each stage's objective. *Stage I (Pre-training)* trains the video and latent dynamics experts on single-view human and robot video with the action expert frozen, $(\lambda_v, \lambda_d, \lambda_a)$=(1.0, 1.0, 0.0). *Stage II (Align Training)* warm-starts from the Stage I checkpoint and unfreezes all experts, training on general robot actions over a 3-view (head + two wrists) canvas, (1.0, 0.5, 1.0) (the action expert and proprioception encoder are (re)initialized rather than loaded). The three views are composed into a single L-shape canvas—the head view spanning the full width across the top (2/3 of the height) and the two wrist views placed side by side along the bottom—so all cameras are modeled jointly within one video stream. *Stage III (Post-training)* fine-tunes all experts on the target embodiment, (1.0, 0.1, 1.0).

All stages optimize with AdamW (β_1=0.9, β_2=0.95, weight decay 0.01) under a cosine learning-rate schedule with peak learning rate 1×10^{-4}, 5% linear warmup, and a minimum learning rate of 1% of the peak; gradients are clipped to a global norm of 1.0. All training is conducted on

64 NVIDIA H20 GPUs. We train in `bf16` mixed precision with DeepSpeed ZeRO-2 sharding, gradient checkpointing, and a fixed seed of 42; the T5 text encoder, the Wan2.2 VAE, and the LDM are frozen throughout. Stage-specific settings (per-GPU batch size, schedule length, and loss weights) are summarized in Tab. 6.

D. Real-World Setup and Task Description

Hardware. We deploy on two real-robot platforms, shown in Fig. 7. The *gripper platform* (top) is a dual-arm AgileX PiPER system, with an AgileX Pika Gripper mounted on each arm. It observes the scene through three cameras: an Intel RealSense D435i on the head and an Intel RealSense D405 on each wrist. Demonstrations are teleoperated with two AgileX Pika Sense handles paired with two AgileX Pika Stations. The *dexterous platform* (bottom) pairs two Tianji arms with two Wuji multi-fingered dexterous hands, and uses an Orbbec Gemini 335L head camera together with an Intel RealSense D405 on each wrist.

We evaluate on seven real-world manipulation tasks that span bimanual and dexterous-hand settings and cover a range of skills, including precise pick-and-place, fine-grained positional adjustment, articulated- and deformable-object manipulation, cross-arm handover, dexterous multi-finger manipulation, and long-horizon multi-step reasoning. Five are executed on the gripper platform and two on the dexterous platform. Detailed instructions are given below.

Gripper Platform

Sorting. *Move the holder to the brown zone, put stationery inside, then put fruit into the basket.* This is a long-horizon task that requires categorizing multiple object types and placing them into their designated regions in the correct order.

Shift Test Tube. *Shift the edge test tube inward by two slots.* This task tests high-precision, small-scale positional control, as the target displacement spans only two slots.

Tidy Desk. *Stow the clutter into the box and flip the lid closed.* Beyond pick-and-place, the task requires articulated-object manipulation to close the lid after all items are stowed.

Fold Shirt. *Fold the cloth.* This task involves deformable-object manipulation, where the configuration of the cloth changes continuously throughout the interaction.

Handover Mug. *Pick up the right cup, hand it over to the left arm, and place it on the left plate.* This is a bimanual task that requires coordinated cross-arm handover followed by accurate placement.

Dexterous Platform

Place Rubik's Cube. *From a clutter of diverse objects, locate the Rubik's cube, pick it up, and place it into the basket.* This task couples target identification among many distractors with a dexterous multi-finger grasp.

Spray Water. *The right hand picks up the spray bottle and passes it to the left hand; the left hand reorients it and passes it back to the right hand, which grips it and presses the trigger to spray water onto the plant.* This is a bimanual dexterous task requiring in-hand tool use with independent finger actuation to operate the trigger.

Table 7: Latent-dynamics regression, full per-rate breakdown (scale-normalized MSE $\downarrow$ on translation+rotation). Columns are sampling rates; best per column in bold. Bottom rows list the train/validation clip counts per rate.

	High-rate			Low-rate		
Method	50 Hz	25 Hz	12.5 Hz	10 Hz	5 Hz	2.5 Hz
LDM	**0.056**	**0.158**	**0.426**	**0.607**	**1.815**	**3.956**
w/o motion align.	0.131	0.487	1.718	2.581	8.497	20.686
w/o multi-stride	0.057	0.170	0.482	0.712	2.205	4.700
w/ pixel recon.	0.063	0.188	0.503	0.742	2.260	5.070
w/ hard VQ	0.093	0.268	0.738	1.064	3.236	7.692
n_{train}	676,447	358,711	196,395	157,868	75,828	31,276
n_{val}	169,413	89,808	49,070	39,489	18,928	7,763

E. Latent Dynamics Regression

We detail the regression probe used to evaluate the latent dynamics. Our goal is to quantify how much action-relevant information the latent dynamics from our LDM already encode. To this end, we keep the latent dynamics fixed and train only a shallow two-layer MLP head to regress the motion from them. Because this head is deliberately shallow, its accuracy reflects the information already in the latent dynamics rather than capacity the head adds.

Data. We build the evaluation set from RoboTwin over all 50 tasks, using $2{,}500$ clean and $2{,}500$ randomized episodes. RoboTwin is not part of the LDM training corpus, so all variants are probed on an unseen domain. We sweep six sampling rates: at stride s, each episode is cut into clips of eight frames taken s apart, with consecutive clips offset by $8s$ frames so that they do not overlap. Near-static transitions are discarded so that they do not dilute the error. The remaining clips are split 8:2 into probe-training and validation; the resulting counts are listed in Tab. 7, ranging from about 0.68M training clips at 50 Hz to 31K at 2.5 Hz.

Regression target. Each 8-frame clip yields seven adjacent transitions. For each, the ground-truth motion is the difference between consecutive absolute end-effector states: a 12-D per-step target stacking the translation (Δp) and rotation (Δq) of both arms. Each modality is divided by a fixed scale (0.01 m for translation, 0.08 rad for rotation), matching the scale used for motion alignment during LDM training.

Head and optimization. Each model's latent is first mapped by a shared $\mathrm{Linear}(D \to 512)$ projection (identity-initialized when $D = 512$) that only removes cross-model dimensionality differences, followed by a two-layer MLP (LN $\to 512 \to 256 \to$ 12, GELU, dropout). All models use this same head and the same optimization: AdamW with learning rate 10^{-3}, weight decay 10^{-4}, batch size 1024, and 200 epochs under a cosine schedule with no early stopping. A separate head is trained per sampling rate.

Rate bands. RoboTwin renders at a native 50 Hz, and each rate is obtained by taking frames at a fixed stride from this source. The six rates are grouped into a *high-rate* band ($50/25/12.5$ Hz, stride $1/2/4$) and a *low-rate* band

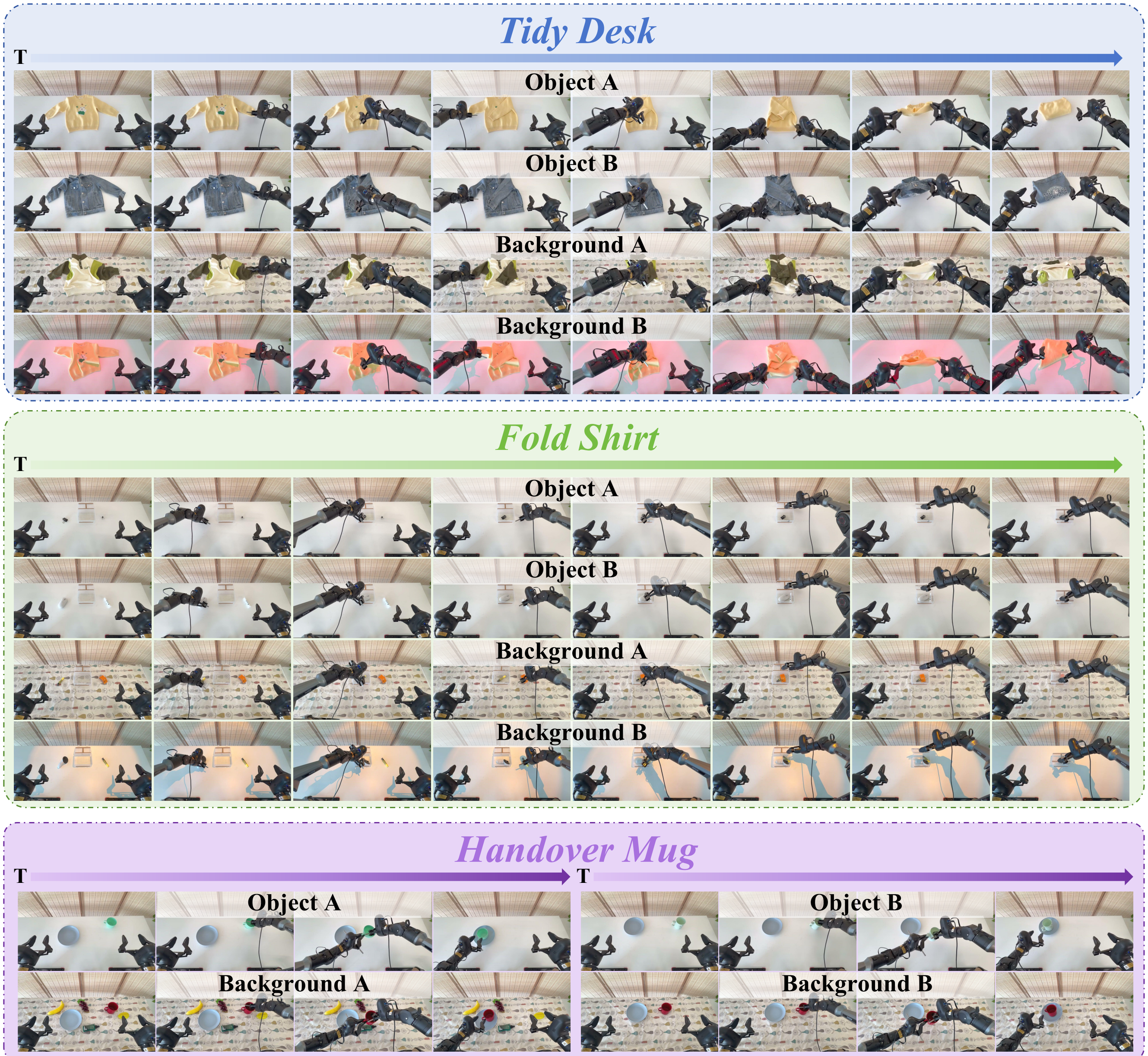

Figure 8: Real-world Generalization Rollouts. Qualitative rollouts of our policy on three real-world tasks, Tidy Desk, Fold Shirt, and Handover Mug. Time proceeds from left to right along the arrow ($\mathbf{T}$). For each task we test robustness under object generalization (Object A / Object B, unseen target instances) and background generalization (Background A / Background B, unseen tabletop textures, clutter, and lighting).

($10/5/2.5$ Hz, stride $5/10/20$), which probe the latent over small and large inter-frame motions, respectively. The full per-rate breakdown is given in Tab. 7.

F. RoboTwin Detailed Results

Tab. 8 reports the per-task success rates on all 50 RoboTwin tasks under both the clean and randomized evaluation settings, complementing the aggregate numbers in the main paper. Our method attains the highest average success rate under both settings (**93.96** clean / **92.78** randomized).

G. More Real-world Deployment Visualizations

Fig. 8 shows additional real-world rollouts of our policy on three tasks, *Tidy Desk*, *Fold Shirt*, and *Handover Mug*. For each task we test generalization to unseen objects and unseen backgrounds. The policy completes every task under all variations, showing robustness to novel objects and scenes.

Table 8: Per-task success rates on RoboTwin under clean and randomized evaluation settings.

Task	**Ours**		Fast-WAM		Lingbot-VA		$\pi_{0.5}$		Motus		LaWAM		ACE-Ego-0	
	Clean	Rand.	Clean	Rand.	Clean	Rand.	Clean	Rand.	Clean	Rand.	Clean	Rand.	Clean	Rand.
Adjust Bottle	**100**	**100**	**100**	**100**	90	94	**100**	99	89	93	**100**	**100**	**100**	**100**
Beat Block Hammer	95	97	**99**	97	96	**98**	96	93	95	88	90	93	98	92
Blocks Ranking RGB	**100**	**100**	**100**	**100**	99	98	92	85	99	97	97	**100**	98	97
Blocks Ranking Size	79	69	**94**	**98**	**94**	96	49	26	75	63	93	89	89	91
Click Alarmclock	98	99	**100**	**100**	99	**100**	98	89	**100**	**100**	**100**	**100**	52	38
Click Bell	**100**	**100**	**100**	**100**	**100**	**100**	99	66	**100**	**100**	**100**	**100**	66	71
Dump Bin Bigbin	99	**97**	97	96	89	96	92	**97**	95	91	97	95	**100**	**97**
Grab Roller	**100**	99	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**
Handover Block	**100**	**89**	95	81	99	78	66	57	86	73	96	87	96	85
Handover Mic	**99**	97	**99**	**100**	94	96	98	97	78	63	93	98	91	94
Hanging Mug	37	49	**58**	**62**	40	28	18	17	38	38	51	43	29	31
Lift Pot	**100**	**100**	**100**	**100**	**100**	99	96	85	96	99	**100**	99	**100**	**100**
Move Can Pot	99	97	90	88	94	97	51	55	34	74	98	93	**100**	**98**
Move Pillbottle Pad	99	**100**	**100**	99	99	99	84	61	93	96	97	90	**100**	**100**
Move Playingcard Away	**100**	**100**	**100**	**100**	**100**	99	96	84	**100**	96	**100**	**100**	**100**	98
Move Stapler Pad	**96**	**89**	77	64	91	79	56	42	83	85	94	87	90	**89**
Open Laptop	99	**100**	98	**100**	92	94	90	96	95	91	**100**	**100**	**100**	98
Open Microwave	79	73	62	45	82	86	34	77	**95**	**91**	41	43	91	85
Pick Diverse Bottles	84	89	80	85	89	82	81	71	90	**91**	**91**	88	84	86
Pick Dual Bottles	99	**100**	**100**	96	**100**	99	93	63	96	90	**100**	95	89	88
Place A2B Left	**99**	95	95	93	97	93	87	82	88	79	98	91	95	**96**
Place A2B Right	**99**	95	93	**99**	97	95	87	84	91	87	89	94	90	94
Place Bread Basket	88	**99**	91	93	**97**	95	77	64	91	94	92	85	92	93
Place Bread Skillet	**99**	**96**	90	93	95	90	85	66	86	83	90	83	94	89
Place Burger Fries	**99**	99	96	99	97	95	94	87	98	98	93	96	98	**100**
Place Can Basket	85	80	71	69	81	**84**	62	62	81	76	**92**	65	78	82
Place Cans Plasticbox	95	97	99	96	**100**	**99**	94	84	98	94	**100**	95	**100**	98
Place Container Plate	99	99	96	**100**	99	97	99	95	98	99	**100**	**100**	98	**100**
Place Dual Shoes	91	**96**	94	88	94	89	75	75	93	87	**98**	94	95	**96**
Place Empty Cup	**100**	**100**	**100**	**100**	**100**	**100**	**100**	99	99	98	99	**100**	**100**	**100**
Place Fan	95	**97**	96	96	**99**	93	87	85	91	87	92	93	94	93
Place Mouse Pad	**98**	**99**	83	89	93	96	60	39	66	68	91	84	96	95
Place Object Basket	**97**	**97**	89	88	91	88	80	76	81	87	92	90	93	89
Place Object Scale	**100**	91	90	**97**	96	95	86	80	88	85	95	88	95	92
Place Object Stand	89	**97**	90	94	**99**	96	91	85	98	**97**	92	93	95	94
Place Phone Stand	93	98	**97**	**99**	**97**	97	81	81	87	86	93	94	91	98
Place Shoe	99	**100**	96	99	98	98	92	93	99	97	**100**	**100**	**100**	**100**
Press Stapler	97	85	90	97	85	82	87	83	93	**98**	**98**	97	**98**	**98**
Put Bottles Dustbin	92	**97**	**95**	90	87	91	84	79	81	79	94	92	94	93
Put Object Cabinet	77	80	**94**	**89**	85	87	80	79	88	71	90	82	82	79
Rotate QRcode	**99**	91	93	89	96	91	89	87	89	73	94	89	94	**95**
Scan Object	95	95	89	92	**96**	91	72	65	67	66	**96**	90	95	**97**
Shake Bottle	**100**	**100**	**100**	**100**	**100**	97	99	97	**100**	97	**100**	**100**	**100**	**100**
Shake Bottle Horizontally	**100**	**100**	**100**	**100**	**100**	99	99	99	**100**	98	**100**	**100**	**100**	**100**
Stack Blocks Three	97	93	95	97	**99**	**98**	91	76	91	95	90	75	87	82
Stack Blocks Two	**100**	**100**	**100**	**100**	**100**	98	97	**100**	**100**	98	**100**	97	**100**	**100**
Stack Bowls Three	87	80	80	81	86	83	77	71	79	**87**	**90**	80	80	85
Stack Bowls Two	99	95	92	98	94	98	95	96	98	98	**100**	**99**	96	98
Stamp Seal	**97**	99	90	94	96	97	79	55	93	92	89	88	94	**100**
Turn Switch	71	45	61	59	44	45	62	54	**84**	**78**	47	56	59	57
Average	**93.96**	**92.78**	91.88	91.78	92.90	91.50	82.74	76.76	88.66	87.02	92.64	89.80	91.12	90.62